

HOW BIG DATA WORLD IS AFFECTING GEOSCIENCE EDUCATION: A STUDY IN HIGH-RANK US-COLLEGES

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Keywords: Geoinformatics, big data world, US-colleges, education

The shift of our world from a traditional, industrial-based society to a knowledge-based, big-data world has proven to touch the lives of people worldwide. Geoinformatics is an emerging field that facilitates this transition within the geoscience community. Geoinformatics refers to the science and technology used to develop information science infrastructure to address problems and interpretations of large-scale geological phenomena. The benefits of geoinformatics are constantly growing as scientific and technological advances in Artificial Intelligence and large data sets queries continue to develop exponentially, yet the education surrounding geoinformatics fails to retain the same pace. A manual search for geoinformatics courses in the course catalogs of U.S. colleges within the best geology departments ranked by U.S. News revealed that these top programs have little to no courses involving geoinformatics. However, in ranking the schools in order of geoinformatics courses offered within their geology courses, it was found that the Western colleges provided more geoinformatics courses than the Eastern colleges with few exceptions. This phenomenon can be understood by looking at the differences in the history, cultural attitudes, and values of the West in comparison to the East. For example, Silicon Valley, a region that serves as a global center for advanced technology and innovation, along with several other big data headquarters, such as Google, Facebook, Linkdn, are located in the Western U.S. Although education surrounding geoinformatics is sparse, the literature review finds that knowledge in the form of geoinformatics is of vital importance and provides benefits in several geo-applications and disciplines. Thus, the U.S. education system should shift focus towards geoinformatics and help it gain prominence within the school science education system in the U.S on all levels. The implications of this research prompt an urgent need for a detailed scholarly analysis of research in this area and the curation of a standardized curriculum for geoinformatics within the U.S. college system. The permeation of geoinformatics through these methods would properly equip our future geologists with the knowledge required for the big data world needs of 2050.

Introduction

Traditionally, the world's economies have been centered around the production and consumption of goods. Towards the mid-to-late 20th century, around the 1970s and 1980s, the Information Age began (Castells, 1997). This period was marked by the rapid advancements and widespread adoption of digital technologies. This has led to unprecedented levels of access to and dissemination of information, as well as profound changes in the way we communicate, work, and live.

A Google Scholar search for the phrase "Geoinformatics" yielded 181,000 results, while a search for "Geoinformatics Education" produced 28,400 hits, primarily from Europe. The literature review revealed that studies on geoinformatics education are mainly located in Europe, with few originating from the United States. Blišťan's study (Blišťan et al., 2015) highlights the importance of geoinformatics education, emphasizing its role in contributing to the development of a knowledge-based society. Despite the United States' reputation as a technological leader, the literature suggests a lack of geoinformatics education in the country. This gap prompted the hypothesis that U.S. universities may not prioritize geoinformatics courses.

Literature Review

Methodology

In this study, the "U.S. News & World Report" 2022 rankings for "Best Geology Programs" were used to determine which colleges to source data from. This specific ranking was used because it's expected for schools that are supposedly the strongest in the Geology field to follow major trends in the geoscience world as well as fulfill the science communities' needs, such as the rise of geoinformatics, in order to stay competitive. A manual search for undergraduate geoinformatics courses within the geology courses offered in the twenty-four U.S. colleges ranked in the list were done. However, five of the colleges were eliminated from the study because they only provided catalogs for specific semesters, rather than a general course catalog for the department. These course catalogs were readily available on the school's websites. Amongst the literature review, there was no established or explicit definition for "geoinformatics". It is important to acknowledge this "grey zone" because it properly addresses the complexity and ambiguity of geoinformatics and highlights the need for careful consideration

and critical thinking to navigate through this field. So, in order to discern the geoinformatics courses from the rest of the geology courses, there were key words and phrases that helped to indicate whether a course is or is not a geoinformatics course using this loose, general definition of geoinformatics: the science and technology of handling, analyzing, interpreting, and communicating geospatial data using computer-based tools and techniques. The different known branches of geoinformatics were also used as a reference, such as remote sensing, geographic information systems (GIS), spatial analysis, and data mining. It is important to note that while many courses may teach skills useful in geoinformatics applications, such as programming, data analysis, and scientific computing, it is not a primarily geoinformatics course. For example, there were many quantitative geology courses that use mathematical methods and computer programming study the Earth's processes but would not qualify as a geoinformatics course because quantitative geology is more focused on understanding the physical and chemical processes that control geological phenomena, while geoinformatics is more focused on the development of tools and methods for managing and analyzing spatial data in a variety of fields.

Results

School Name	# of Geology Courses Offered	# of Geoinformatics Courses Offered	%
University of Minnesota - Twin Cities	59	1	1.7
University of Texas - Austin	81	2	2.5
Stanford University (lowest % in the West-USA)	34	1	2.9
Pennsylvania State University - University Park	83	3	3.6
Yale University	53	2	3.8
Indiana University - Bloomington	46	2	4.3
University of Colorado - Boulder	65	3	4.6
Rice University	70	4	5.7
University of California - Santa Barbara	86	5	5.8
University of California - Los Angeles	60	4	6.7
Massachusetts Institute of Technology	42	3	7.1
Colorado School of Mines	41	3	7.3
California Institute of Technology	41	3	7.3
Ohio State University	67	5	7.5
University of Wisconsin - Madison	51	4	7.8
University of California - Berkeley	36	3	8.3
University of Washington	42	4	9.5
Columbia University (highest % in the East-USA)	30	4	13.3

[% 0-4 red; 5-6 yellow; 7+ green]

Discussion

Executing a search for the exact phrase “Geoinformatics” into “Google Scholar” resulted in about 181,000 searches and another search for the phrase “Geoinformatics” with “Education” resulted in about 28,400 searches. A closer look at the searches on “Geoinformatics Education” showed studies mainly located in Europe and not many stemming from the U.S. For example, in “The Importance Of Geographic Information Systems Education At Universities In The Process Of Building A European Knowledge- Based Society”, Blišťan emphasizes the significance of a geoinformatics education in universities to contribute towards the development of a knowledge-based society (Blišťan, et. al., 2015). He highlights the various applications and benefits, and specifically mentions the establishment of a successful geoinformatics education program at the Technical University of Kosice in Slovakia. The literature review for information on a geoinformatics education in the U.S. was concerning because the U.S. is known to be one of the most technologically advanced countries for several decades, so it is expected for the U.S. to be constantly evolving with the technological advancements as well. However, it seems that the U.S. is lacking in the geoinformatics field due to the literature review. This prompted the hypothesis that there would be insignificant findings in whether universities in the U.S. offered geoinformatics courses. The results of the study confirmed this hypothesis. In viewing the course catalogs of the colleges with the best geology departments, the most geoinformatics courses offered within a geology program was five and the least being one. The highest percentage of geoinformatics courses to geology courses offered was 13%. This was an expected result, but in further looking through the data, a pattern was found. The data revealed that schools in the West seemed to offer more geoinformatics courses than schools in the East with some exceptions. This phenomenon can possibly be because the differences in the cultural attitudes and values of the West and the East. It is important to note that this is a broad generalization and should not be taken as both regions are diverse and multifaceted. However, it can be seen that the Western U.S. can be seen more entrepreneurial, innovative, and forward-thinking, while the eastern U.S. can be seen as more cautious, conservative, and traditional. This may be due to historical patterns, such as the settlement history of the regions and the legacy of different industries and social movements within the regions. For example, the eastern U.S. has a longer history of settlement than the western U.S., which has allowed for the development of more established traditions and cultural norms. This also means it is home to many of the country’s oldest and most established

cultural institutions, such as museums and historical landmarks, which may contribute to a greater emphasis on the preservation of traditions and norms. Additionally, the industries within the East are more traditional, such as finance, law, and manufacturing, which can be seen by the renowned Wall Street and the Financial District. These industries tend to be more conservative and cautious in their approach to business. On the other hand, the Western U.S. has a history of being a technological hub with an innovative culture due to Silicon Valley. Silicon Valley, located in Northern California, began its emergence as a technology hub as a result of the growth of the semiconductor industry in the region due to the establishment of Fairchild Semiconductor in 1957 followed by the founding of Intel in 1968 (Sturgeon, 2000). During the 1970s and 1980s, the growth of the personal computer industry in the region fueled by the success of companies like Apple attracted venture capital firms, such as Sequoia Capital, which provided funding and support for startups in the area (Sturgeon, 2000). Finally, in the 1990s and 2000s, the growth of the internet and rise of Facebook and Google further solidified the Valley's position as a global technological hub (Sturgeon, 2000). These industries provide a culture of innovation that values risk-taking and forward-thinking. This presence of leading technology companies along with access to venture capitals provides the regions, thus the schools within the region, a network of resources, expertise, funding, and talent required to keep up with the constantly evolving big-data world.

*Possibility: demographics -> eastern has an older population which may also contribute to a greater emphasis on tradition and caution [Population Distribution by Age | KFF](#)

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Discussion:

(In the intro: only a sentence about what europe does, and another sentence about that we should track other continents on this Educational need of geoinformatics to keep the pace or outrun them.)

Methodology:

Why those universities? Following the rank..because ranking shows that those universities are following the major trends of societal and sci-communities' needs

Results:

Eastern: less geoi courses with exception of Columbia
Western: more geoi courses with exception of Stanford
geoi/total courses % shows the same pattern

Figures:

- 1.Flowchart of your methodology
- 2.GIS percentages geoinfcours/total courses
- 3.GIS No of geoinfcourses

Tables:

Questions: